ChemPubSoc Europe

DOI: 10.1002/cphc.201402382

Averaging Techniques for Reaction Barriers in QM/MM Simulations**

April M. Cooper[a] and Johannes Kästner*[b]

Herein, we test and compare different techniques to obtain averaged reaction barriers from quantum mechanics/molecular mechanics (QM/MM) simulations based on snapshots taken from molecular dynamics. Reasonable values can be obtained from a fairly small sample of well-chosen snapshots if an exponential averaging, also called Boltzmann averaging, is used. Snapshots with geometries close to the expected transition state are to be picked preferentially. Exponential averaging, arithmetic averaging, and simply taking the minimum barrier are compared to free-energy calculations from umbrella sampling. Three reactions within a protein in a water environment are used as test cases.

1. Introduction

QM/MM simulations, the combination of quantum mechanics (QM) and empirical force fields (MM, molecular mechanics),[1,2] have become an established method to study biochemical processes.[3–6] Its advantages are obvious: the active site of an enzyme, the area in which chemical bonds are broken or formed, is described by quantum mechanics. The enzyme environment, which influences the active site while itself being chemically inert, is handled on the force-field level. This saves computational effort while ensuring an appropriate description of whole enzymatic systems and their aqueous environment.

While the computational cost for each energy evaluation is kept at bay by the QM/MM separation, enzymes are still large systems with many thousand degrees of freedom. The study of reaction mechanisms is typically based on geometry optimizations, often starting from the crystal structure. These, however, normally lead to local minima when systems as large as proteins are treated. A rigorous alternative is the investigation of free-energy surfaces, which are accessible by molecular dynamics (MD) or Monte Carlo sampling. A typical geometry optimization requires 10^2 to 10^3 energy evaluations, proper sampling needs $>10^6$ such steps. This is prohibitively expensive for many QM/MM treatments. However, it should be noted that some MD studies using QM/MM expressions based on density functional theory were reported.[7–11]In an alternative to QM/MM, namely, cluster models, the enzyme environment is truncated at some distance of the reactive center.[12] Typically, the whole model is described with a quantum chemical method. Atoms close to the truncation are frozen. Again, this restricts the system to the configuration of the crystal structure, which might miss out important flexibility of the protein.

In the simulation of molecular reactions in the gas phase, one can often cover all possible conformers of the reactant state and investigate the associated transition states. This full configurational sampling is impossible in a QM/MM setting due to the system size.

In recent years, it has been realized that a single set of geometry optimizations, and thus a single reaction profile, often leads to highly biased results. A combination of the limiting cases of full free-energy sampling and local geometry optimization was sought for. Nowadays the accepted approach is to initially perform an MD simulation of the reactant state in a pure force-field description. Snapshots, that is, instantaneous geometries at several points in time, are taken from the resulting MD trajectory as starting points for several geometry optimizations. Barriers $\Delta^{\ddagger}E_i$ are calculated for each of the N resulting minima, which are subsequently averaged to result in one mean barrier. While this approach is increasingly used in biomolecular simulations of enzymatic processes, to our knowledge its accuracy was never thoroughly investigated. Different possibilities exist for averaging barriers. Should a straightforward arithmetic mean [Eq. (1)]:

$$\Delta^{\ddagger}E_{\mathrm{arithm}} = \frac{1}{N}\sum_{i=1}^{N}\Delta^{\ddagger}E_i \qquad (1)$$

between the individual barriers be calculated, as has been done several times,[13–18] sometimes with additional higher-accuracy energy calculations?[19–21] One could also weigh the barriers according to the relative potential energies of the reactant states E^{RS} [Eq. (2)]:

[a] A. M. Cooper
Computational Biochemistry Group
Institute of Theoretical Chemistry
University of Stuttgart, Pfaffenwaldring 55
70569 Stuttgart (Germany)

[b] Prof. Dr. J. Kästner
Computational Biochemistry Group
Institute of Theoretical Chemistry
University of Stuttgart, Pfaffenwaldring 55
70569 Stuttgart (Germany)
E-mail: kaestner@theochem.uni-stuttgart.de

[**] QM/MM: Quantum Mechanics/Molecular Mechanics

 © 2014 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

$$\Delta^{\ddagger}E_{\text{weight}} = \frac{\sum_{i=1}^{N} exp(-E_i^{RS}/k_BT)\Delta^{\ddagger}E_i}{\sum_{i=1}^{N} exp(-E_i^{RS}/k_BT)} \quad (2)$$

Should the energy difference between the lowest-energy reactant minimum and the lowest-energy transition state be used? Alternatively, the minimum of the reaction barriers [Eq. (3)]:

$$\Delta^{\ddagger}E_{\text{min}} = min\Delta^{\ddagger}E_i \quad (3)$$

has been chosen,[22] since in transition state theory this path would provide the highest rate and could be deemed dominating. An increasingly popular choice is an exponential average [Eq. (4)]:

$$\Delta^{\ddagger}E_{\text{exp}} = -k_BT\,ln\left(\frac{1}{N}\sum_{i=1}^{N} exp(-\Delta^{\ddagger}E_i/k_BT)\right) \quad (4)$$

with k_B being Boltzmann's constant and T being the absolute temperature. To our knowledge, this was first used by Logunov and Schulten in 1996[23] and later on independently by several groups.[17,18,20,24–26] $\Delta^{\ddagger}E_{\text{exp}}$ is smaller than $\Delta^{\ddagger}E_{\text{arithm}}$, it approaches $\Delta^{\ddagger}E_{\text{arithm}}$ for large T values and $\Delta^{\ddagger}E_{\text{min}}$ for small ones. According to Jarzynski's equation,[27] the free energy ΔA of a process equals the exponential average of the work of a series of paths drawn from a canonical ensemble. However, while the initial structures of QM/MM optimizations are drawn from a canonic ensemble, $\Delta^{\ddagger}E_i$ are not. Nevertheless, $\Delta^{\ddagger}E_{\text{exp}}$ may serve as an approximation to the free-energy barrier of the reaction $\Delta^{\ddagger}A$. More pictorially, one can argue that if there are several possible reaction paths, the one with the lowest reaction barrier will be the most favored one. If the differences in the barrier heights $\Delta^{\ddagger}E_i$ get sufficiently large (larger than k_BT), then the reaction path that has the smallest barrier will dominate the reaction.

Obviously, rigorous approaches to obtain $\Delta^{\ddagger}A$ require full sampling of the configurational space. While this generally is too costly for QM/MM with density functional theory or higher QM methods, it can be achieved with either semiempiric QM methods or the fit of the QM potential energy surface by the empirical valence bond (EVB)[28] method or the related multiconfiguration molecular mechanics (MCMM) approach.[29] Such treatments, however, are out of the scope of the present article.

The present paper sets out to provide and test a stable protocol for averaging snapshots by comparison to full free-energy calculations based on umbrella sampling[30,31] and the calculation of average potential energy barriers from many snapshots by different ways of averaging.

2. Calculational Details

While we aim at investigating averaging techniques for QM/MM simulations, here we study transitions with pure force-field calculations to allow for complete sampling and for performing a large number of geometry optimizations of snapshots. We calculated up to 100 snapshots in three different reactions, many more than would typically be used in QM/MM calculations.

We used peptidyl prolyl isomerase A (also called cyclophilin A), which catalyzes the *cis*/*trans* isomerization of proline peptide bonds. The structure of the protein is shown in Figure 1. This *cis*/*trans* isomerization as well as a change in the ring puckering of the proline substrate (backwards and forwards) were studied, see Figure 2.



Figure 1. Schematic representation of the protein peptidyl prolyl isomerase A with substrate. The proline residue of the substrate is represented by balls and sticks.

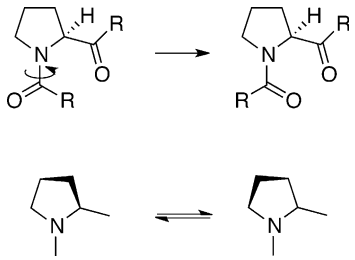


Figure 2. *Cis*/*trans* isomerization of the proline peptide bond (top) as well as the change in the ring puckering of proline (bottom).

The protein coordinates were taken from the protein data base (entry 1RMH),[32] described using the Chemistry at Harvard Molecular Mechanics (CHARMM) force field,[33] and protonated and solvated in a periodic box of TIP3P water[34] using Visual Molecular Dynamics (VMD) version 1.9.[35] This resulted in a system of 15257 atoms. Molecular dynamics simulations were performed with the Nanoscale Molecular Dynamics (NAMD) code version 2.8.[36] The temperature was kept at 300 K by a Langevin thermostat; a time step of 1 fs was chosen. All bond lengths involving hydrogen atoms were constrained. For the protein equilibration, cycles of minimization of 2000 steps followed by MD runs of 50000 steps were per-

© 2014 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

formed, where the protein atoms were restrained (restrain parameters were lowered in each cycle: infinity—frozen protein atoms—, 5.0, 2.0, and 0.1 kcal mol^{-1} Å^{-2}). This protocol allows water molecules to penetrate cavities within the protein. Then, a free MD run of 4 ns was carried out; the Langevin piston Nosé–Hoover method was used to keep the system at 300 K and 1 bar. The same settings were used for the umbrella sampling reference calculations, see below.

Snapshots were taken from that MD trajectory at regular intervals of 5 ps. For the following geometry optimizations, which are to mimic QM/MM optimizations, a truncated system was used. All protein atoms and any water molecules with at least one atom within 15 Å from the substrate were retained, all other water molecules were deleted. This reduced the number of atoms from 15257 to about 4600 depending on the snapshot. Whole residues (amino acids or water molecules) with at least one atom within 8 Å of the proline N atom were optimized (about 360 atoms), all other atoms were frozen to their positions from the MD. This is the same protocol as we typically employ in QM/MM investigations of reaction mechanisms.[24,37–39]

All geometry optimizations were performed in ChemShell[40] by using the DL-FIND optimization library[41] with default convergence criteria. Transition states were located by a superlinearly converging variant[42] dimer method.[43,44]

The free-energy profile from umbrella sampling[30,31] simulations with umbrella integration analysis[45] served as a reference. The reaction coordinates in all three reactions were torsion angles. Force constants of 0.03 kcal mol^{-1} deg^{-2} were used, except for the region around the transition state in the *cis/trans* reaction, where 0.05 kcal mol^{-1} deg^{-2} were used. 37 windows were sampled for the *cis/trans* reaction and 17 windows for the ring-puckering rearrangement, with 0.1 ns of sampling per window.

3. Results

Among the three transitions studied, the *cis/trans* isomerization of proline is the most relevant for comparison with real QM/MM simulations. It has a free-energy barrier of (75.9 ± 1.0) kJ mol^{-1}, see Table 1, which is similar to many enzymatic reactions. Therefore, the main discussion is focused on this system. The conclusions drawn from the other two reactions are similar, though. The ring puckering has a forward barrier of 10.7 kJ mol^{-1} and a backward barrier of 11.8 kJ mol^{-1} in the free energy, which is small enough to be overcome several times during an MD of 5 ns.

65 snapshots were used for the analysis of the *cis/trans* isomerization. The distribution of their potential-energy barriers $\Delta^{\ddagger}E_i$ is shown in Figure 3. The arithmetic mean $\Delta^{\ddagger}E_{\text{arithm}}$ is with

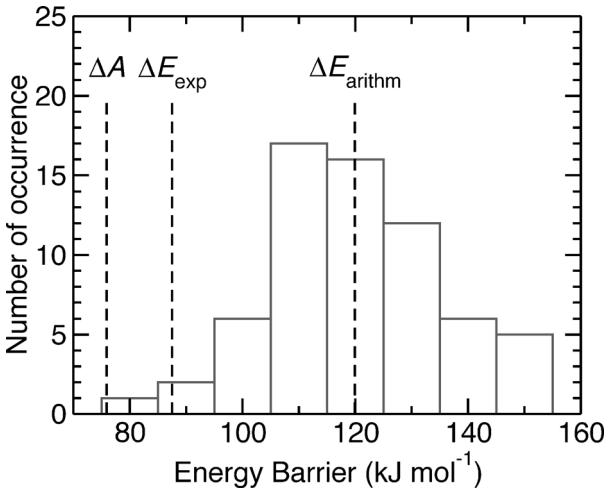


Figure 3. Histogram of $\Delta^{\ddagger}E_i$ of the *cis/trans* rearrangement of proline with the different estimations of the barrier indicated.

119.9 kJ mol^{-1}, significantly higher than the free-energy barrier from umbrella sampling, 75.9 ± 1 kJ mol^{-1}, see Table 1. The weighted barrier according to Equation (2) is with 113.2 kJ mol^{-1} only a little closer. By contrast, the exponentially averaged barrier, $\Delta^{\ddagger}E_{\text{exp}} = 87.5$ kJ mol^{-1} is much closer to the reference value. One snapshot resulted in a particularly small barrier of 77.1 kJ mol^{-1}, which is quite close to the sampled free-energy barrier. This single snapshot dominates the exponential average.

As expected, the exponential average is closer to the free-energy barrier than the arithmetic average. However, in a real QM/MM calculation it is infeasible to optimize 65 snapshots. Typically, about 3–10 are used. If these are chosen randomly, the result may differ significantly. The left graph in Figure 4 shows the exponential average of randomly chosen snapshots. It is clear that any outcome between 80 and over 140 kJ mol^{-1} is possible if only a few snapshots are used. However, it has been suggested[17] to actively pick those snapshots for which a small barrier $\Delta^{\ddagger}E_i$ can be expected from the value of the reaction coordinate in the MD structure. Small values of $\Delta^{\ddagger}E_i$ dominate $\Delta^{\ddagger}E_{\text{exp}}$, so they are the most important contributions to calculate. Several additional larger barriers would not contribute much to the sum in Equation (4), but would increase N. However, ten high-energy snapshots neglected in that way lead to an underestimation of the barrier of only $k_B T \ln 10 \approx 5.7$ kJ mol^{-1} at 300 K, 100 neglected ones to only 11.5 kJ mol^{-1}. In the light of the accuracy of typical DFT calculations, and since incomplete sampling generally leads to an overestimation in the first place, this error is acceptable. The lower panel of Figure 3 shows the exponential average of the snapshots sorted by the torsion angle in the MD structure

Table 1. Comparison of different estimates of barrier heights (all energies in kJ mol^{-1}).

	Cis/trans	Forward	Backward
ΔA	75.9 ± 1.0	10.7 ± 0.4	11.8 ± 0.3
$\Delta^{\ddagger}E_{\text{arithm}}$	119.9	16.1	14.6
$\Delta^{\ddagger}E_{\text{min}}$	87.5	12.7	13.2
$\Delta^{\ddagger}E_{\text{exp}}$	77.1	7.2	9.7
$\Delta^{\ddagger}E_{\text{weight}}$	113.2	23.3	12.8
gas phase	42.0	–	11.7
solvated X-ray	102.6	–	14.6

© 2014 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

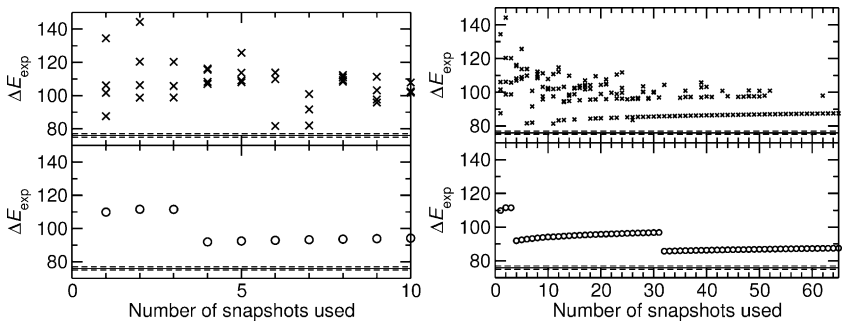


Figure 4. Top: exponential average of randomly chosen snapshots. Bottom: snapshots sorted by initial torsion angle. ΔA from umbrella sampling is indicated by a solid horizontal line, its confidence interval by dashed lines. The left graph focuses on small numbers of snapshots as typically used in QM/MM simulation. The right graph shows the averages over all snapshots calculated.

before the optimization. A high torsion angle suggests a snapshot with the environment of the active site pre-arranged to favor the transition state. Indeed, the second-lowest barrier was found for the fourth snapshot when the latter are sorted by torsion angle. The lowest barrier is found much later at snapshot 32, which explains the step in the lower left panel of Figure 3. Since small barriers dominate the exponential average, the circles in the lower panel of Figure 3 show low values as soon as the one low barrier is included. In such an approach, four snapshots would be sufficient to achieve a reasonable average.

The ring-puckering reactions lead to qualitatively similar results. The barriers are rather small compared to k_BT, so that the exponential average and the arithmetic one are closer. This is less typical for QM/MM calculations. Furthermore, the reaction is quite decoupled from the environment. It is clear from Table 1 that the exponential average is, again, closer to the reference value $\Delta^{\ddagger}A$ than the arithmetic average if all snapshots are used. However, the more relevant comparison to QM/MM is to use only few well-chosen snapshots. Again, MD structures with a reaction coordinate close to the expected transition state are chosen preferentially. Picking only a few snapshots already results in exponential averages close to $\Delta^{\ddagger}A$. Actually, for 2–10 of such snapshots, $\Delta^{\ddagger}E_{exp}$ lies within an interval only twice as wide as the 95% confidence interval (details not shown). However, since the small barrier causes $\Delta^{\ddagger}E_{arithm}$ to be close to $\Delta^{\ddagger}E_{exp}$, also the arithmetic average of few well-chosen snapshots is a good approximation of $\Delta^{\ddagger}A$.

Obviously, one could also choose the minimum barrier as an estimate of $\Delta^{\ddagger}A$. In the *cis/trans* rearrangement, $\Delta^{\ddagger}E_{min}$ is the closest estimate to the reference value. However, this agreement is fortuitous. It cannot be advisable to approximate an average by just a single simulation as this approach is not robust. The same holds for $\Delta^{\ddagger}E_{weight}$: because of the large fluctuations in the reactant state energy, one single snapshot, the one with the lowest reactant state energy, obtains a weight $> 99.99\%$ when calculating $\Delta^{\ddagger}E_{weight}$ while the others are ignored. It is clear from Table 1 that this leads to rather arbitrary results: in the case of the backward reaction, $\Delta^{\ddagger}E_{weight}$ is fortuitously close to the reference value. It is high but still reasonable for the *cis/trans* reaction, but quite unrealistically high for the forward reaction. The energy of the reactant is an inappropriate measure of the importance of the snapshot for the overall reactivity.

A much simpler approach than running MD simulations and taking snapshots is to seemingly rely on experimental data and just solvate the crystal structure and investigate the reaction mechanism starting from that. However, crystal packing is a rather non-natural condition for an enzyme, even though several enzymes can actually perform catalysis if the crystals are soaked with substrate. Furthermore, the crystal structure is an average over an ensemble and not one particularly likely geometry. Fewer water molecules are resolved in the crystal structure than are actually present in a protein. Consequently, and not surprisingly, the calculation of the energy barrier by just solvating the crystal structure and then performing the same geometry optimization as in each MD snapshot results in rather poor approximations to the reaction barrier, see Table 1. Such an approach is not recommended for QM/MM simulations.

4. Discussion

Only in $\Delta^{\ddagger}E_{weight}$, the energy of the reactant is used additionally to the value of the barrier. In principle, a reactant with lower total energy is more likely to occur and should, therefore, play a larger role in the reaction. However, the total potential energy of a system is an extensive quantity, it depends on the system size, in contrast to the size-independent barrier. Also the fluctuations in the total energy depend on the system size. Figure 5 shows the fluctuations of the reactant state energy for the cis/trans reaction. The fluctuations are in the range of 10^3 kJ mol^{-1}, at least an order of magnitude larger than the

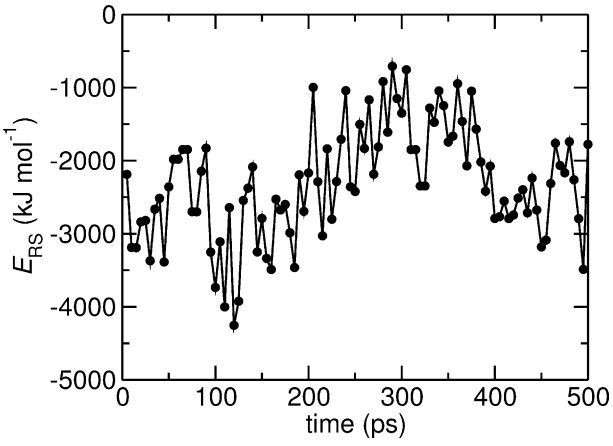


Figure 5. Total potential energy of the reactant state E_{RS} of the snapshots of the *cis/trans* isomerization. Note the huge fluctuation.

© 2014 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

energy barrier and two orders larger than the fluctuations of the barrier. Note that all of the snapshots have been drawn in constant intervals from a rather short molecular dynamics simulation. Thus, the system probably has remained in the basins of these local minima for similar times. Consequently, all of these minima are approximately equally likely. The large fluctuations result in quite extreme weights in $\Delta^{\ddagger}E_{\text{weight}}$. In all three examples studied, one single snapshot completely dominates $\Delta^{\ddagger}E_{\text{weight}}$ with a weight of $> 99.99\,\%$. In the case of the forward reaction of the ring puckering, this is even the snapshot with the second highest barrier. Overall, $\Delta^{\ddagger}E_{\text{weight}}$ picks one almost random snapshot out of the set rather than performing a suitable average. Furthermore, the vibrational energy in the transition mode is more relevant for the probability of a transition to occur than the total energy which is distributed over all modes.

None of the techniques to estimate energy barriers discussed above approach the reference value from umbrella sampling, not even for a very large number of snapshots. This is no big surprise, however, since they use a different model. In the umbrella sampling, like in the real system, the whole enzyme and its solvent in the periodic box are free to move and rearrange. By contrast in the QM/MM simulations, we have to decrease the system size by ignoring most of the environment and freezing most atoms which remain. However, it is clear that our protocol still allows the atoms most relevant for the reaction to move and, thus, accommodate the structural changes induced by the reaction. One could perform QM/MM calculations with the full solvent environment in a periodic setting. However, most standard quantum chemical codes cannot handle periodicity in the electrostatics. Because of this—and due to further technical reasons—we normally restrict ourselves to truncated QM/MM models. Localized reactions that do not involve large-scale rearrangements of the whole protein, such as the ones discussed here, are well-suited for the approach of reducing the number of degrees of freedom in the geometry optimization by freezing a large part of the model. These typically also show only a negligible contribution of the entropy to the chemical step.[46] However, other cases like the ribosome discussed previously[24] require much larger active regions. The modeling requires careful attention to the properties of the system to maintain the predictive power of the method.

5. Conclusions

We have compared different techniques to obtain averaged reaction barriers from QM/MM simulations. They were mimicked by force-field calculations to allow obtaining reference values from full sampling of the system. It is clear that an exponential average (Boltzmann average) of only a few barriers obtained from MD snapshots is sufficient to provide a reasonable estimate of the barrier, as long as these snapshots are well-chosen. An appropriate criterion is to pick snapshots from the MD, which adopt a configuration that is already close to the transition state. In these cases, the environment is in a geometry that favors the transition and results in a small barrier. Alternatives, like the arithmetic average of the barriers, were found to be a worse approximation. The total energies of the reactant minima are irrelevant for the averaging.

Acknowledgements

The authors thank the German Research Foundation (DFG) for financial support of the project within the Cluster of Excellence in Simulation Technology (EXC 310) at the University of Stuttgart.

Keywords: computational chemistry · molecular mechanics · proteins · quantum mechanics · reaction barriers

[1] A. Warshel, M. Karplus, *J. Am. Chem. Soc.* **1972**, *94*, 5612.
[2] A. Warshel, M. Levitt, *J. Mol. Biol.* **1976**, *103*, 227.
[3] D. Riccardi, P. Schaefer, Y. Yang, H. Yu, N. Ghosh, X. Prat-Resina, P. König, G. Li, D. Xu, H. Guo, M. Elstner, Q. Cui, *J. Phys. Chem. B* **2006**, *110*, 6458.
[4] H. Lin, D. G. Truhlar, *Theor. Chem. Acc.* **2007**, *117*, 185.
[5] H. M. Senn, W. Thiel, *Top. Curr. Chem.* **2007**, *268*, 173 – 290.
[6] H. M. Senn, W. Thiel, *Angew. Chem.* **2009**, *121*, 1220; *Angew. Chem. Int. Ed.* **2009**, *48*, 1198.
[7] T. K. Woo, P. M. Margl, P. E. Blöchl, T. Ziegler, *J. Phys. Chem. B* **1997**, *101*, 7877 – 7880.
[8] M. Strajbl, G. Hong, A. Warshel, *J. Phys. Chem. B* **2002**, *106*, 13333 – 13343.
[9] A. Crespo, M. A. Martí, D. A. Estrin, A. E. Roitberg, *J. Am. Chem. Soc.* **2005**, *127*, 6940 – 6941.
[10] B. Ensing, M. De Vivo, Z. Liu, P. Moore, M. L. Klein, *Acc. Chem. Res.* **2006**, *39*, 73 – 81.
[11] Z. Ke, S. Wang, D. Xie, Y. Zhang, *J. Phys. Chem. B* **2009**, *113*, 16705 – 16710.
[12] P. E. Siegbahn, F. Himo, *WIREs Comput. Mol. Sci.* **2011**, *1*, 323 – 336.
[13] H. M. Senn, S. Thiel, W. Thiel, *J. Chem. Theory Comput.* **2005**, *1*, 494 – 505.
[14] S. Cohen, S. Kozuch, C. Hazan, S. Shaik, *J. Am. Chem. Soc.* **2006**, *128*, 11028 – 11029.
[15] P. Hu, Y. Zhang, *J. Am. Chem. Soc.* **2006**, *128*, 1272 – 1278.
[16] R. A. Mata, H.-J. Werner, S. Thiel, W. Thiel, *J. Chem. Phys.* **2008**, *128*, 025104.
[17] R. Lonsdale, J. N. Harvey, A. J. Mulholland, *J. Phys. Chem. B* **2010**, *114*, 1156 – 1162.
[18] C. Z. Christov, A. Lodola, T. G. Karabencheva-Christova, S. Wan, P. V. Coveney, A. J. Mulholland, *Biophys. J.* **2013**, *104*, L5 – L7.
[19] Y. Zhang, J. Kua, J. A. McCammon, *J. Phys. Chem. B* **2003**, *107*, 4459 – 4463.
[20] M. W. van der Kamp, J. Zurek, F. R. Manby, J. N. Harvey, A. J. Mulholland, *J. Phys. Chem. B* **2010**, *114*, 11303 – 11314.
[21] J. Jitonnom, V. S. Lee, P. Nimmanpipug, H. A. Rowlands, A. J. Mulholland, *Biochemistry* **2011**, *50*, 4697 – 4711.
[22] J. Jiang, J. Lu, D. Lu, Z. Liang, L. Li, S. Ouyang, X. Kong, H. Jiang, B. Shen, C. Luo, *PLoS One* **2012**, *7*, e36660.
[23] I. Logunov, K. Schulten, *J. Am. Chem. Soc.* **1996**, *118*, 9727 – 9735.
[24] J. Kästner, P. Sherwood, *Mol. Phys.* **2010**, *108*, 293.
[25] R. Lonsdale, K. T. Houghton, J. Zurek, C. M. Bathelt, N. Foloppe, M. J. de Groot, J. N. Harvey, A. J. Mulholland, *J. Am. Chem. Soc.* **2013**, *135*, 8001 – 8015.
[26] R. Lonsdale, S. Hoyle, D. T. Grey, L. Ridder, A. J. Mulholland, *Biochemistry* **2012**, *51*, 1774 – 1786.
[27] C. Jarzynski, *Phys. Rev. Lett.* **1997**, *78*, 2690.
[28] J. Åqvist, A. Warshel, *Chem. Rev.* **1993**, *93*, 2523.
[29] M. Higashi, D. G. Truhlar, *J. Chem. Theory Comput.* **2008**, *4*, 790 – 803.
[30] G. M. Torrie, J. P. Valleau, *Chem. Phys. Lett.* **1974**, *28*, 578 – 581.
[31] G. M. Torrie, J. P. Valleau, *J. Comput. Phys.* **1977**, *23*, 187 – 199.
[32] Y. Zhao, H. Ke, *Biochemistry* **1996**, *35*, 7356 – 7361.
[33] A. D. MacKerell, Jr., D. Bashford, R. L. Bellott, R. L. Dunbrack Jr., J. D. Evanseck, M. J. Field, S. Fischer, J. Gao, H. Guo, S. Ha, D. Joseph-McCarthy, L. Kuchnir, K. Kuczera, F. T. K. Lau, C. Mattos, S. Michnick, T. Ngo,

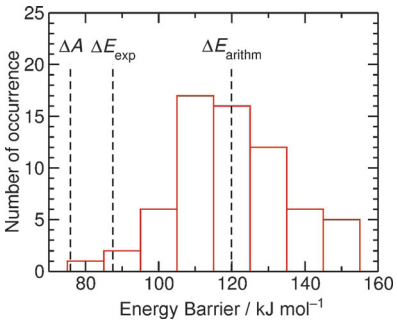
© 2014 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

D. T. Nguyen, B. Prodhom, W. E. Reiher III, B. Roux, M. Schlenkrich, J. C. Smith, R. Stote, J. Straub, M. Watanabe, J. Wiorkiewicz-Kuczera, D. Yin, M. Karplus, *J. Phys. Chem. B* **1998**, *102*, 3586.
[34] W. L. Jorgensen, J. Chandrasekhar, J. D. Madura, R. W. Impey, M. L. Klein, *J. Chem. Phys.* **1983**, *79*, 926.
[35] W. Humphrey, A. Dalke, K. Schulten, *J. Mol. Graphics* **1996**, *14*, 33.
[36] J. C. Phillips, R. Braun, W. Wang, J. Gumbart, E. Tajkhorshid, E. Villa, C. Chipot, R. D. Skeel, L. Kalé, K. Schulten, *J. Comput. Chem.* **2005**, *26*, 1781.
[37] J. B. Rommel, J. Kästner, *J. Am. Chem. Soc.* **2011**, *133*, 10195.
[38] E. Abad, R. K. Zenn, J. Kästner, *J. Phys. Chem. B* **2013**, *117*, 14238 – 14246.
[39] E. Abad, J. B. Rommel, J. Kästner, *J. Biol. Chem.* **2014**, *289*, 13726 – 13738.
[40] S. Metz, J. Kästner, A. A. Sokol, T. W. Keal, P. Sherwood, *Comput. Mol. Sci.* **2014**, *4*, 101.
[41] J. Kästner, J. M. Carr, T. W. Keal, W. Thiel, A. Wander, P. Sherwood, *J. Phys. Chem. A* **2009**, *113*, 11856.
[42] J. Kästner, P. Sherwood, *J. Chem. Phys.* **2008**, *128*, 014106.
[43] G. Henkelman, H. Jónsson, *J. Chem. Phys.* **1999**, *111*, 7010.
[44] R. A. Olsen, G. J. Kroes, G. Henkelman, A. Arnaldsson, H. J onsson, *J. Chem. Phys.* **2004**, *121*, 9776.
[45] J. Kästner, W. Thiel, *J. Chem. Phys.* **2005**, *123*, 144104.
[46] H. M. Senn, J. Kästner, J. Breidung, W. Thiel, *Can. J. Chem.* **2009**, *87*, 1322.

Received: June 2, 2014
Revised: July 29, 2014
Published online on ■■ ■■, 2014

© 2014 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

Different methods to average reaction barriers from quantum mechanics/molecular mechanics (QM/MM) simulations are compared. It is advisable to take snapshots from a molecular dynamics simulation rather than directly using the static crystal structure as a basis. The exponential average of the barriers obtained from the snapshots is recommended.



*A. M. Cooper, J. Kästner**

■■ – ■■

Averaging Techniques for Reaction Barriers in QM/MM Simulations

© 2014 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim